

STROOP TEST

PSY310: Lab in Psychology

September 7, 2026

Eva Joseph | AU2420030

GitHub Link: <https://github.com/evaj-beep/PSY310-LAB-IN-PSYCH.git>

Introduction

The Stroop Color and Word Test (SCWT) is a neuropsychological test extensively used for both experimental and clinical purposes. It assesses the ability to inhibit cognitive interference, which occurs when the processing of a stimulus feature affects the simultaneous processing of another attribute of the same stimulus (Stroop, 1935).

The most common iteration of the SCWT, requires the participants/respondents to read three separate tables as quickly as possible. Two of them depict a "congruous" condition in which participants are expected to read color-words printed in black ink (W) and name distinct color patches, whereas in the third table, the color-word (CW) condition, color-words are printed in an inconsistent color ink (for example, "red" is printed in green ink). In this "incongruent" condition, participants must name the color of the ink rather than read the word.

Through this experiment, the participants are required to perform a less automated task (i.e., naming ink color) while inhibiting the interference arising from a more automated task (i.e., reading the word; MacLeod and Dunbar, 1988; Ivnik et al., 1996).

This difficulty in inhibiting the more automated process is called the Stroop effect (Stroop, 1935).

Using this paradigm, we can determine an individual's cognitive processing speed, attentional capacity and their level of cognitive control. This test provides a brief, yet penetrating glance into the human mind and behaviors that are otherwise very inherent in our interaction with the world (Farnsworth, 2019).

Method

Participants

The study involved a convenience sample of one male university student. The participant was 20 years old. Participants were recruited through personal peer contacts at Ahmedabad University. Prior to participation, the subject provided informed consent, and all acquired data was anonymised to safeguard confidentiality.

Materials

- **Hardware:** Stimuli were presented on a Lenovo IdeaPad Gaming laptop equipped with an integrated display monitor and keyboard. Participant responses were recorded via the built-in keyboard, using the 'r' key to indicate a red font color and the 'b' key to indicate a blue font color.
- **Software:** The task was programmed, presented, and recorded using PsychoPy (2026 version). PsychoPy controlled the presentation timing of each stimulus, randomized trial presentation, and automatically logged both response accuracy and response latency (reaction time) for each trial.
- **Stimuli:** Visual stimuli consisted of the color-name text strings "RED" and "BLUE" created directly within the PsychoPy timeline. Text was displayed centrally on the

screen using exclusively red and blue font colors across two experimental conditions:

- Congruent: The target word matched its display font color (e.g., "RED" presented in red font, "BLUE" presented in blue font).
- Incongruent: The target word differed from its display font color (e.g., "RED" presented in blue font, "BLUE" presented in red font).

Procedure

Before the experiment began, the participants' consent was obtained after which they were briefed on the task. The experiment had no in-built instructions, so I made sure the participants knew exactly what task they had to do and what keys needed to be pressed for the right responses.

As the task began, the participants were presented with the words and were told to identify if they were congruent in respect to their ink colour or incongruent. There were four conditions/scenarios that they were presented with.

SCENARIO 1: The word RED was written in RED ink

SCENARIO 2; The word RED was written in BLUE ink

SCENARIO 3: The word BLUE was written in RED ink

SCENARIO 4; The word BLUE was written in BLUE ink

When the words matched the ink, participants were instructed to push the "r" and "b" keys, respectively. In other words, they were instructed to use the "r" key in scenarios 1 and 3 and the "b" key in scenarios 2 and 4. Their job was to disregard the word's actual meaning and focus only on the ink used to express it. As a result, the task had four conditions and four parameters.

Results

For Participant 1, reaction time and response accuracy were investigated in the congruent and incongruent Stroop trials. For Participant 1, the mean reaction time was 0.447 s (SD = 0.141) for congruent trials and 0.574 s (SD = 0.250) for incongruent trials as illustrated in Table 1.

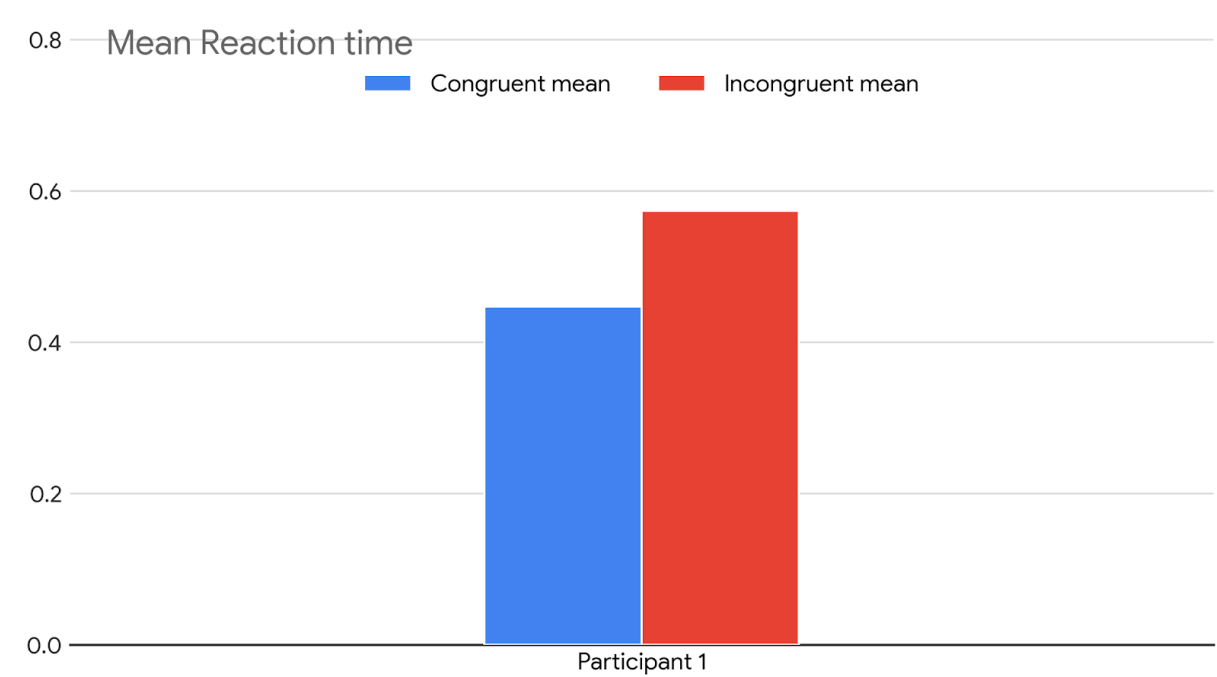
In both the congruent and incongruent conditions, Participant 1 achieved an accuracy of 100%. The accuracy of the participant in the congruent condition and the incongruent condition are presented in Table 1.

The mean difference in reaction times between incongruent and congruent trials (i.e., Stroop interference score) for Participant 1 was 0.127 s (126.67 ms). The incongruent condition had higher mean reaction times compared to the congruent condition.

Participant	Condition	Mean RT (s)	SD (s)	Accuracy

1	Congruent	0.447	0.141	100%
1	Incongruent	0.574	0.250	100%

Figure 1. Participant 1: Mean reaction time (s) for the congruent and incongruent Stroop trials.



Participant 1: Mean reaction time (s) for the congruent and incongruent Stroop trials.

Discussion

The data clearly show the standard Stroop interference effect, as predicted. With an average cognitive processing delay of 126.67 ms, the participant was significantly slower on incongruent trials (Mean = 573.50) than on congruent trials (Mean = 446.83). Rather than a speed-accuracy trade-off or unintentional answer errors, perfect accuracy (100%) in both cases suggests that this latency difference represents true cognitive interference during trial selection.

These results are consistent with well-established cognitive psychology research on executive control and selective attention (Stroop, 1935; MacLeod, 1991). Because reading is a highly automated, involuntary process, automatic word reading interferes with visual color recognition. Executive control mechanisms must actively inhibit the dominant reading response in order to process the target font color when word meaning and font color clash (for

example, the word "BLUE" presented in red font), resulting in quantifiable reaction time latencies.

These results show how PsychoPy accurately records visual-cognitive response latencies in real time.

References

Scarpina, F., & Tagini, S. (2017). The Stroop Color and Word Test. *Frontiers in psychology*, 8, 557. <https://doi.org/10.3389/fpsyg.2017.00557>

Farnsworth, B. (2019, November 7). *The Stroop effect – How it works and why it has a profound impact*. iMotions. <https://imotions.com/blog/learning/research-fundamentals/the-stroop-effect/>